HB+

Google Authentication Portal

IT Intern Assignment | Project Overview & Evaluator Guide

Repository	github.com/Shivprasadpravindongapure/hb-
Live web app	jolly-entremet-88c205.netlify.app/login
Platforms	Web, Android, iOS (single React Native codebase)
Core services	Google OAuth 2.0, Firebase Authentication, Cloud Firestore

Project summary

HBPlus Auth is a cross-platform team portal that signs users in with Google and checks their email domain before granting dashboard access. It uses React Native and React Native Web for a shared interface, Firebase Authentication for identity, and Firestore for user profiles and database access rules. The application includes a login screen, protected dashboard, sign-out, loading and error states, and a responsive layout.

Assignment coverage

Requirement	Implementation
Google sign-in	Firebase Google provider; web uses Firebase popup sign-in, while native uses Expo AuthSession and Firebase credentials.
Domain restriction	Shared exact-match email validator normalizes case and checks the configured allowed-domain list. Firebase auth state signs out accounts outside that list.
Protected dashboard	Navigation renders the dashboard only after authentication and authorization have resolved. The protected-route component blocks content while loading or redirecting.
Logout, loading, errors	Auth context exposes sign-in/sign-out, displays pending state, and reports authentication errors in the login UI.
Database and roles	Firestore profile documents store user identity, domain, timestamps, and role. New users receive the default user role; rules prevent clients from changing restricted fields.
Cross-platform structure	Expo, React Native, React Native Web, React Navigation, and TypeScript support a shared app for web and mobile.

How the application works

Sign-in and authorization flow

1. Start

Firebase auth listener restores any existing session. A loading screen prevents protected content from flashing before session status is known.

2. Sign in

The login screen launches Google sign-in. Web uses Firebase signInWithPopup; Android/iOS use Expo AuthSession and then exchange the Google credential with Firebase.

3. Validate

AuthContext receives the Firebase user and passes the email through the exact-domain validator. Unlisted domains are signed out and shown an access-restricted error.

4. Load profile

Authorized users have their Firestore profile created or updated. Existing roles are retained; profile loading can fall back to Firebase Auth profile data if Firestore is unavailable.

5. Protect and sign out

The navigator selects the dashboard only for an authorized session. Sign out clears the Firebase session and returns the user to login.

Security layers

- Client-side domain check provides immediate user feedback and prevents an unauthorized account from remaining signed in through the app.
- Firestore rules validate signed-in identity, email domain, email verification, and document ownership. Creation rules require the default user role; updates cannot alter role, UID, email, or domain.
- OAuth client secrets are not required in the web bundle. EXPO_PUBLIC_* values are public build-time configuration; Firebase rules remain necessary to protect database data.

Repository structure

Path	Responsibility
src/services/authService.ts	Platform-specific Google authentication and Firebase sign-out.
src/context/AuthContext.tsx	Central authentication state, domain enforcement, profile load, loading and errors.
src/utils/domainValidator.ts	Reusable exact email-domain validation.
src/navigation/ and src/components/	Navigation, protected route, shared auth UI and loading/error components.
src/screens/	Login and authenticated dashboard screens.
firestore.rules / firebase.json	Firestore database authorization and Firebase CLI rules deployment configuration.
netlify.toml / README.md	Web export, SPA routing/security headers, and setup/deployment guide.

Run, configure, and evaluate

Local setup

Requirements: Node.js 20 or newer, npm, and a Firebase project with Google Authentication enabled. Configure the Firebase web app values and Google OAuth client IDs using the template in `.env.example`. Do not commit `.env` or OAuth client secrets.

```
git clone https://github.com/Shivprasadpravindongapure/hb-.git
cd hb-
npm install --legacy-peer-deps
copy .env.example .env    # then fill in the Firebase/OAuth values
npm run web
```

Useful project commands

`npm run web` starts the web app. `npm run android` and `npm run ios` start the native platforms. `npm run type-check` validates TypeScript. `npm run build:web` exports the production web bundle.

Production deployment

- Netlify builds the web app with `npx expo export --platform web` and publishes `dist`; the Netlify config includes SPA fallback routing.
- Add the deployed hostname to Firebase Authentication authorized domains and configure the Google OAuth authorized web origin.
- Set the build-time `EXPO_PUBLIC_ALLOWED_EMAIL_DOMAINS` variable to the intended allowlist. Redeploy after changing a build-time variable.
- Deploy database rules with `npx firebase-tools@latest deploy --only firestore:rules --project hbplus-authentication-93ca5` (authenticated Firebase project access required).

Important: assignment behavior versus temporary Gmail testing

The assignment?s acceptance example requires **@hbplus.fit to be allowed and @gmail.com to be denied**. The repository was subsequently configured to support a comma-separated allowlist, and its current documented production test setting includes both `hbplus.fit,gmail.com`. Therefore Gmail is allowed when both domains are present. For an evaluator run that must match the assignment exactly, set Netlify?s `EXPO_PUBLIC_ALLOWED_EMAIL_DOMAINS` to `hbplus.fit` only, redeploy, and deploy Firestore rules that likewise allow only `hbplus.fit`.

Suggested evaluator test matrix

Test	Expected result
Sign in with employee@hbplus.fit	Google authentication succeeds and the authorized dashboard is shown.
Sign in with a Gmail account (strict assignment configuration)	Access-restricted error; user is signed out and dashboard is not accessible.
Sign in with another unlisted domain	Access denied by the exact-match allowlist.
Open dashboard while signed out / refresh session	Unauthenticated user is routed to login; valid Firebase session is restored after refresh.
Sign out	Session is cleared and login screen is shown.

Validation run during project work: TypeScript check passed and Expo web production export completed. Google sign-in still requires the evaluator/browser to complete the account-selection interaction.

Thank you for reviewing the HBPlus Auth project.